Speaking App PRD — Student Form

> **How to use this form (작성 방법)**
>
> 1. Copy this whole form into **markdown.kr**.
> 2. For each item, read the **Tip** and the **Example** (from the sample *Order Up! — Café Talk*), then write your own answer under ✏️ **My answer**.
> 3. Delete the Tip and Example lines when you finish, so only your answers remain.
> 4. Do **not** edit the section **"Technical Requirements"** at the end. Paste it into AI Studio as it is.
> 5. Save as `.md` and submit to the LMS with your Stitch screen captures.

- **Name / Student ID:** 이가낙/2023315617
- **My situation (상황):** *(e.g., ordering at a café, asking for directions, introducing yourself, shopping, at the doctor's)*

ordering at the cafe

1. App Name (앱 이름)

Tip: Short and easy to remember. Add a subtitle that shows the situation.

Example: Order Up! — Café Talk

✏️ **My answer:**

SpeakEasy — Café Talk

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2. Why We Are Building It (왜 만드는가)

Tip: Write (1) the problem students have, (2) how the app solves it, and (3) one learning goal that starts with *"Students can say..."*.
Use the word **intelligibly** (not "perfectly") in your goal.

Example:

- Problem: Students can read café-ordering expressions but rarely say them out loud.
- Solution: Students repeat **Listen → Speak → Get feedback → Try again** on their own.
- Goal: Students can say five café-ordering expressions intelligibly.

🖊 **My answer:**

- Problem: Beginner English learners often feel shy when ordering food or drinks in English. They may understand the sentences but have difficulty saying them aloud.
- Solution: The app lets students listen to a café conversation, practice speaking, and get friendly feedback.
- Goal: Students can say ...

  Students can say five café-ordering expressions intelligibly and complete a short dialogue.

3. Target Users (대상 사용자)

Tip: Include grade, **CEFR level**, one emotional characteristic, one common mistake, and when/where they use the app.

Example:

- Grade: Middle school 1st–2nd year
- Level: CEFR A1–A2
- Characteristic: Shy about speaking; afraid of making mistakes
- Common mistake: Dropping small words like *a* and *please*
- Use context: On a smartphone, 5–10 minutes, for review or homework

🖊 **My answer:**

- Grade: High school and university students, ages 16–22
- Level: CEFR A1-A2
- Characteristic: Shy about speaking and afraid of making mistakes
- Common mistake: Forgetting small words such as “a” and polite words such as “please”
- Use context: On a smartphone or laptop, for 3–5 minutes per session, for self-study or English class practice

4. Screen Layout (화면 구성)

Tip: Look at your **Stitch screens** and describe each one using **Top / Middle / Bottom**. Write button names in [brackets].
The target sentence must be the **largest text**. The mic button needs **three states** (Ready → Listening → Done).

Example (Practice Screen):

- Top: Mia's question in a speech bubble + Korean meaning
- Middle: Target sentence (largest text) + [🔊 Listen] [🐢 Slow]
- Bottom: Large round [🎤 Mic] button (Ready → Listening → Done), feedback box, [Try Again] [Next →] [Skip]

🖊️ **My answer:**

Screen 1: Start Screen

- Top: App title “SpeakEasy — Café Talk” and a small café illustration.
- Middle: A situation card saying “You are ordering at a café.” Show the goal: “Practice 5 useful sentences.”
- Bottom: [Start Practice] button and the notice: “Your voice is processed to turn it into text and is not recorded or saved.”

Screen 2: Practice Screen

- Top: The café partner’s question in a speech bubble, with its Korean meaning.
- Middle: The target English sentence in the largest text. Include [Listen] and [Slow] buttons, and show “Sentence 1 of 5.”
- Bottom: A large round [Mic] button, a feedback box, and [Try Again], [Next], and [Skip] buttons.
- Mic button states: Ready → Listening → Done. Disable the mic button while model audio is playing. Show a guidance message if speech recognition is unsupported or microphone permission is denied.

Screen 3: Result Screen

- Top: “Café Talk Complete!” and a supportive message.
- Middle: Show the number of sentences practiced out of 5 and whether the dialogue is completed. Do not show numeric pronunciation scores or percentages.
- Bottom: [Practice Again] and [Back to Start] buttons.

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5. Key Features (주요 기능)

Tip: Describe the flow for **one sentence** in order. Keep the six steps below and fill in the details for your app.

Example:

1. See the situation — The partner's question appears and is read aloud once.
2. Listen — [🔊 Listen] plays the target sentence; [🐢 Slow] plays it slowly.
3. Speak — Tap [🎤]; listening stops after about 1.5 seconds of silence.
4. Compare — Compare word by word; ignore capitalization and punctuation.
5. Feedback — The character gives short feedback (Section 7).
6. Try again / Next — Students can retry, go next, or skip.

🖊️ **My answer:**

1. See the situation — The café partner's question appears on screen and is read aloud once.
2. Listen — [Listen] plays the target sentence. [Slow] plays it at a slower speed.
3. Speak — The student taps [Mic] and says the target sentence. Listening stops after about 1.5 seconds of silence when supported by the browser.
4. Compare — The recognized words appear as text. Capitalization and punctuation differences are ignored. Success depends on recognizing key words, not native-like pronunciation.
5. Feedback — The partner character gives short, friendly feedback based on the rules in Section 7.
6. Try again / Next — The student can retry, move to the next sentence, or skip. After all five turns, the app opens the Result Screen.

After all sentences are finished, the app moves to the Result Screen.

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6. Practice Dialogue (연습 대화)

Tip:

- Write **5 turns**. The AI must use your sentences **exactly as written**.
- Match the sentences to the **CEFR level** in Section 3. Keep student sentences short.
- Mark 1–3 **key words** per sentence. If the student says all key words, it counts as success.

Example:

No.	Partner	Student target sentence	Korean meaning	Key words
1	Hi! What can I get for you?	Can I have a hot chocolate, please?	핫초코 한 잔 주시겠어요?	have, hot, chocolate
2	What size would you like?	A small one, please.	작은 걸로 주세요.	small

🖊️ **My answer:**

- **Partner character (name / role):**

No.	Partner	Student Target Sentence	Korean Meaning	Key Words
1	Hi! What can I get for you?	Can I have a hot chocolate, please?	핫초코 한 잔 주시겠어요?	have, hot, chocolate
2	What size would you like?	A small one, please.	작은 걸로 주세요.	small, one, please
3	Would you like anything to eat?	Can I have a cookie, too?	쿠키도 하나 주시겠어요?	have, cookie, too
4	Is that everything?	Yes, that's everything.	네, 이상입니다.	yes, everything
5	That will be five dollars.	Here you are. Thank you!	여기 있습니다. 감사합니다!	here, thank, you

Partner Character

Mia, a friendly café worker who speaks clearly and patiently.

Star Rules

★★★ Finished all five turns and tried speaking on each turn.
★★ Finished three or four turns.
★ Finished the dialogue or practiced at least one turn.

Stars reward participation, not pronunciation accuracy.

7. Feedback Rules (피드백 규칙)

Tip: This is the most important section. Use the theories from the Week 7 lesson.

- Praise first, **one** thing to fix, **two sentences or fewer**, simple English + Korean in parentheses.
- No "wrong," no number or % scores, no descriptions of tongue/lip/mouth positions.
- Judge by **"Did the partner understand?"**, not native-like pronunciation.
- Change the feedback by attempt: **1st recast → 2nd elicitation (first-letter hint) → 3rd chunking**.

Example:

Situation	Feedback
Success	Perfect! I understood your order clearly. (주문을 정확하게 알아들었어요!)
1st failure (recast)	Nice try! You can say, "Can I have a hot chocolate, please?" (이렇게 말해 볼까요?)
2nd failure (elicitation)	You're almost there! "Can I have a hot chocolate, p___?" (마지막 단어는 p로 시작해요.)

Character personality / speaking style	Mia is friendly, patient, and encouraging. She uses simple English and a short Korean explanation when helpful.
My rules	1. Praise the student's effort first. 2. Give one small, clear suggestion at a time. Use two sentences or fewer. 3. Never say "wrong," give numeric or percentage scores, or describe tongue, lip, or mouth positions. 4. Judge whether the partner understands the key message, not whether the speech sounds native-like. 5. Change support by attempt: 1st attempt = recast; 2nd attempt = elicitation with a first-letter hint; 3rd attempt = chunking.
Situation	**Feedback (English + Korean)**
Success	"Perfect! I understood your order clearly." (주문 내용을 잘 이해했어요!)
Success + small word missing	"Nice job! Add 'please' to sound more polite." (잘했어요! 더 공손하게 'please'를 붙여 보세요.)
1st failure (recast)	"Nice try! You can say, 'Can I have a hot chocolate, please?'" (좋은 시도예요! 이렇게 말해 보세요.)
2nd failure (elicitation)	"You're almost there! 'Can I have a hot chocolate, p...?'" (거의 다 왔어요! 마지막 단어는 p로 시작해요.)
3rd failure (chunking)	"Let's say it in small parts: 'Can I have' / 'a hot chocolate' / 'please.'" (짧은 부분으로 나누어 말해 봐요.)
No speech recognized	"I didn't catch that. Try speaking a little louder, or tap Listen first." (잘 들리지 않았어요. 조금 더 크게 말하거나 먼저 듣기를 눌러 보세요.)

8. Do Not (하지 말 것)

Tip: AI often adds features you did not ask for. Write what the app must **not** do in three areas. The privacy items below are **required** — keep them and add your own.

✏️ **My answer:**

Privacy and Safety (required — keep these)

- Do not save or send audio recordings.
- Do not collect personal information (name, student ID, school, email). No login or sign-up.
- No ads, external links, or payment features.
- No photos of real people or famous characters.

Learning Design (add at least 2)

Privacy and Safety (required — keep these)	• Do not save or send audio recordings. • Do not collect personal information such as name, student ID, school, or email. No login or sign-up. • No ads, external links, or payment features. • No photos of real people or famous characters.
Learning Design	• Do not shame students or use “wrong” as feedback. • Do not judge success by native-like pronunciation. • Focus on whether the partner understands the key message. • Do not show numeric pronunciation scores or percentages.
Other (optional)	• Do not add unrelated games or extra features that make a 3–5 minute practice session longer.

9. Completion Checklist (완성 체크리스트)

Tip: Write sentences you can **check with your eyes** while testing as different students (a good student, a student who makes a mistake, a student who says nothing). You will use this list in Part 3.
"The app is good" ✗ → "The mic button changes to 'Listening' when tapped" ✓

🖊️ My answer:

No.	Checklist Item	Completed
1	The Start Screen shows the café situation and privacy notice.	☑
2	The app contains five dialogue turns with the target sentences from Section 6.	☑
3	Each turn shows the partner’s question, Korean meaning, target sentence, and key words.	☑
4	The [Listen] button reads the target sentence in English.	☑
5	The [Slow] button reads the sentence more slowly.	☑
6	The mic button visibly changes from “Ready” to “Listening” to “Done.”	☑
7	The mic button is disabled while model audio is playing.	☑
8	The recognized speech appears as text.	☑
9	Capitalization and punctuation differences are not counted as failures.	☑
10	The first, second, and third failed attempts use recast, elicitation, and chunking in that order.	☑
11	Saying nothing shows a “Let’s try again” message.	☑
12	The student can retry, go next, or skip a sentence.	☑
13	The Result Screen appears after all five turns.	☑
14	The result shows practice completion without numeric pronunciation scores.	☑
15	Audio is not recorded or saved, and no personal information is collected.	☑
16	Unsupported speech recognition, denied microphone permission, and in-app browser use show a guidance message.	☑
17	The app can complete the full practice without an AI service being available.	☑

- **Speech recognition:** Browser built-in Web Speech API (SpeechRecognition), language `en-US`
- **Read-aloud:** Browser built-in SpeechSynthesis, language `en-US`, normal rate 0.9 / slow rate 0.7
- **Feedback sentences:** Gemini, following Section 7. If Gemini does not respond, show the matching example sentence from the Section 7 table. The app must work to the end without AI.
- Do **not** run SpeechRecognition and MediaRecorder at the same time. Use speech recognition only.
- Disable the mic button while the model audio is playing.
- Show a guidance message when: the browser does not support speech recognition ("Please open it in Chrome."), microphone permission is denied, or the app is opened in an in-app browser such as KakaoTalk or Instagram.
- Show a small notice on the Start Screen: "Your voice is processed to turn it into text and is not recorded or saved."
- The deployed address must use **https** for the microphone to work.

Before You Submit (제출 전 자기 점검)

- [ ] Section 3 includes a **CEFR level**, and the sentences in Section 6 match it.
- [ ] Section 4 uses **Top / Middle / Bottom** and matches my Stitch screens.
- [ ] Section 6 has **5 turns** with key words.
- [ ] Section 7 has **recast → elicitation → chunking** and no % scores.
- [ ] Section 9 has sentences I can check by testing.
- [ ] My partner checked this PRD, and I revised it.
- [ ] I recorded the AI tools and main prompts in my AI use log (.md).